Project Motivation

I chose the CelebA dataset because it presents an important problem in machine learning, where models learn the wrong thing. In this dataset, the majority of blond-haired people are female, so a model trained normally picks up on this pattern and learns to associate being female with having blond hair, rather than actually looking at the hair. This means the model works well on average, but performs poorly on blond males, who appear rarely in the training data that the model never really learns what they look like. I chose this problem because I was interested in figuring out how I could solve this myself given that the bias comes from a simple imbalance in the data. I wanted to see if a straightforward fix could close the gap without needing a complex method.

Dataset Details

The dataset used in this project is CelebA; a large collection of 202,599 celebrity face images each labeled with 40 binary attributes such as hair color, gender, age, and facial features. The dataset is split into 162,770 training images, 19,867 validation images, and 19,962 test images. The prediction task is whether the person has blond hair or not. Gender is not given to the model at any point during training, but it is used at evaluation time to split the test results into four groups of blond female, blond male, non-blond female, and non-blond male so we can measure how well the model performs on each one separately. The key imbalance in the data is that there are around 22,000 blond female training images but only around 1,400 blond male training images, meaning blond males make up less than 1% of all training data. This imbalance is what causes a normally trained model to fail on blond males since it sees them so rarely that it never learns to recognize them reliably.

Method

I used a ResNet-18 convolutional neural network pretrained on ImageNet as the base model for this project. ResNet-18 is an established image classification architecture, and starting from ImageNet weights means it already knows how to detect low level visual features like edges, colors, and textures before any training on CelebA begins. I replaced the final classification layer with a new two class output layer for blond versus not blond, and fine tuned the entire network on CelebA. To address the bias, I identified all blond male images in the training set and repeated their indices 30 times in the training data, so that during each training epoch the model encounters blond males roughly as often as blond females. The idea is that if the model sees enough blond males, it can no longer rely on gender as a shortcut and is forced to learn actual hair color features instead. To prevent the model from memorizing the repeated images, I applied random data augmentation during training including random cropping, horizontal flipping, and brightness and contrast adjustments. I initially trained the model using SGD, but found that it caused the blond male accuracy to drop significantly in later epochs, likely because SGD caused the model to overfit on the repeated minority images. Switching to the Adam optimizer stabilized training and produced consistent results. The model was trained for 10 epochs using Adam with a fixed learning rate of 0.003. I chose oversampling because it directly targets the data imbalance. Another advantage of this approach is that it is computationally inexpensive because the backbone is already pretrained on ImageNet, there is no need to train a large network from scratch, and the oversampling itself adds no extra computation since it just changes which images are sampled during training, rather than adding any new model components.

Results

```
Final Results:
  Non-blond women: 92.24%
  Non-blond men:   91.67%
  Blond women:     91.81%
  Blond men:       92.22%
  Overall average: 91.99%
  Worst group:     91.67%
Average accuracy target met.
Blond men accuracy target met.
```

```
Baseline Final Results:
  Non-blond women: 93.51%
  Non-blond men:   95.94%
  Blond women:     87.62%
  Blond men:       52.22%
  Overall average: 82.32%
  Worst group:     52.22%
Average accuracy target not met.
Blond men accuracy target not met.
```

Discussion

The oversampling approach successfully met both targets, achieving 91.99% average accuracy and 92.22% accuracy on the blond male minority group. A baseline model trained on the same data without any intervention achieved only 52.22% accuracy on blond males despite reaching 95.94% on non-blond men, confirming that the bias is real. The gap between the baseline's 52.22% and the oversampling model's 92.22% on blond males demonstrates that simply showing the model more minority examples during training was enough to stop it from relying on gender as a shortcut and force it to learn actual hair color features. The overall average also improved from 82.32% to 91.99%. That said, oversampling has limitations since it works by repeating existing images rather than creating genuinely new ones, so with a small enough minority group the model could still partially overfit to those specific faces. With more time I would experiment with higher oversample ratios and longer training to see how far the worst-group accuracy can be pushed.

Code and Data Access

Code: https://github.com/JacobMatti/CSE475-Honors-Contract

Dataset: https://www.kaggle.com/datasets/jessicali9530/celeba-dataset